

Dear Hiring Team,

I am writing to apply for the Data Scientist position at J.D. Power / Autodata Division. With a Master of Science in Computer Science from the University of Calgary and hands-on experience in Python, Pandas-style data processing, SQL, machine learning, and large-scale data pipelines, I am excited by the opportunity to contribute to predictive data products that are accurate, transparent, and useful to real customers.

What stands out to me about this role is the combination of data science, production responsibility, and clear business impact. The posting describes a team that is accountable for every data point it outputs, builds processes that directly affect core valuation products, and explains technical methodology to internal stakeholders and clients. That matches the kind of work I enjoy most: careful data work where modelling, QA, documentation, and communication all matter.

My graduate research focused on building scalable machine learning and data-management systems for complex Earth observation datasets. I developed Python and C++ workflows for data ingestion, cleaning, transformation, storage, retrieval, model training, and evaluation. In one project, I built deep-learning models for digital elevation model super-resolution and evaluated them with statistical and domain-specific metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. A major part of that work was identifying when validation methods produced overly optimistic results and designing more reliable region-based evaluation. That experience maps closely to J.D. Power's emphasis on trustworthy forecasts, QA, and explaining predictive products.

My second major research area involved a DGGS-based framework for managing spatiotemporal satellite imagery. I converted large geospatial inputs into structured tensor datasets, designed retrieval workflows for time-series analysis, reduced storage requirements by 38% on a Canada-wide dataset, and improved processing speed through multithreaded implementation. These projects strengthened my practical understanding of ETL-style pipelines, reproducibility, performance tradeoffs, and the discipline needed to keep large analytical workflows reliable as data changes.

I also bring backend and stakeholder-facing experience. As a back-end developer, I built Django REST APIs backed by PostgreSQL and implemented production-facing logic for authentication, service plans, and payment records. As a teaching assistant at the University of Calgary, I supported students in big data applications, data-at-scale workflows, databases, and analytical coursework, which helped me become comfortable explaining technical work to both technical and non-technical audiences.

I would bring strong curiosity, attention to detail, and a practical engineering mindset to this role. I have experience with Python, SQL, Git-based development, cloud-data coursework and mentoring, machine learning evaluation, and AI workflow automation, and I would be excited to deepen my production AWS, Tableau, and automotive market data experience within J.D. Power's data science team.

Thank you for considering my application. I would welcome the opportunity to discuss how my data science, software development, and applied research background can contribute to J.D. Power's valuation and predictive analytics products.

Sincerely,  
Amir Mirzai Golpayegani  
amirgolpaa24@gmail.com